

Merrivoss Bioanalysis Inc

ST03 growth-signal program: rapid hERG/DDI screen of 38 compounds

Final Report

Study MV-2015-266 (LIMS M15/BIO/06146)

Client: Selverant

Study lead: Dr M. Rinalvesi, DMPK Lead

Report date: December 28, 2015

1 Summary

38 compounds from the Selverant aminoquinazoline series 03 were tested for Selverant in the IKr inhibition, Five-isoform DDI screen. Testing started on December 8, 2015 and results were reported on December 28, 2015. All accepted runs met matrix acceptance criteria; one run was rejected and repeated (see Section 6). 1 of 38 compounds had a hERG IC₅₀ below 1 μ M. Most compounds (35 of 38) inhibited hERG with IC₅₀ below 10 μ M, so margins will depend on free therapeutic exposure. Inhibition above 50% at 10 μ M was most frequent for CYP2C19.

Table 1 Study summary

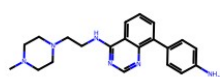
Metric	Value
Project	MV-2015-266
Client	Selverant
Compounds tested	38
Series / chemotype	Selverant aminoquinazoline series 03 (Aminoquinazoline)
Assays	hERG patch clamp, CYP inhibition panel (1A2/2C9/2C19/2D6/3A4) at 10 μ M
Turnaround (calendar days)	20
Price	USD 18,700
hERG IC ₅₀ <1 μ M	1
hERG IC ₅₀ 1-10 μ M	34
hERG IC ₅₀ $\geq$ 10 μ M	3
hERG median IC ₅₀ (μ M)	3.8
E-4031 control IC ₅₀ (nM)	11
CYP1A2 >50% inhibition @10 μ M	10
CYP2C9 >50% inhibition @10 μ M	8
CYP2C19 >50% inhibition @10 μ M	13
CYP2D6 >50% inhibition @10 μ M	4
CYP3A4 >50% inhibition @10 μ M	10
Compounds with any CYP >50%	17
Ketoconazole (3A4) IC ₅₀ (μ M)	0.027

2 Objective

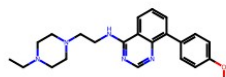
To rank the test article set for IKr inhibition, Five-isoform DDI screen liability so that Sclerant could choose analogues for progression.

3 Test items

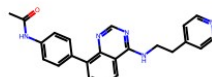
38 compounds were received as 10 mM DMSO solutions in one accession lot on December 8, 2015 and logged against M15/BIO/06146. Identity and concentration were taken as supplied by the client. Stocks were stored at -20 °C and thawed once for testing.



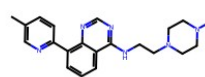
ST03-01210



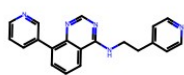
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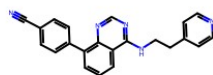
ST03-01224



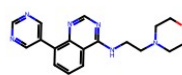
ST03-01228



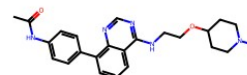
ST03-01230



ST03-01242



ST03-01246



ST03-01252

4 Methods

IKr inhibition

CHO cells stably expressing hERG (KCNH2) were recorded in whole-cell configuration on a 16-channel automated patch clamp platform at room temperature. Cells are held at -80 mV, depolarized to +40 mV for 2 s and repolarized to -50 mV for 2 s to elicit the tail current, at 0.1 Hz. Each compound was applied cumulatively at 0.12, 0.37, 1.1, 3.3, 10 and 30 μ M (0.3% DMSO final). Each concentration is held until the tail-current block reaches a steady state (change below 5% over the last three sweeps), with a minimum of 3 and a maximum of 8 minutes; compounds still drifting at 8 minutes are flagged as slow-onset.

E-4031 is run as a separate 6-point concentration response (1-300 nM) on every run.

- Matrix acceptance: seal resistance ≥ 500 M Ω before break-in and ≥ 200 M Ω throughout
- Tail current ≥ 200 pA at baseline; run-down $\leq 10\%$ over the vehicle period
- E-4031 IC₅₀ within 5-50 nM; runs outside this range are rejected and repeated
- IC₅₀ fitted per cell (four-parameter logistic); the compound IC₅₀ is the geometric mean of finite cell IC₅₀s when these are the majority of accepted cells, otherwise reported as >30 μ M
- Where precipitation is seen at 30 μ M, that concentration is excluded and the highest evaluable concentration is reported

Five-isoform DDI screen

Pooled human liver microsomes (0.1 mg/mL protein) were incubated with each compound at 10 μ M and an isoform-selective probe substrate at approximately K_m , in duplicate, for 10 minutes at 37 °C after NADPH addition. Metabolite formation is measured by LC-MS/MS against vehicle (0.1% DMSO) incubations; the lower limit of quantitation of each metabolite is at least 20-fold below the vehicle signal.

Probe substrates and controls

Isoform	Probe substrate	Metabolite	Positive control
CYP1A2	Phenacetin	Acetaminophen	Furafylline
CYP2C9	Diclofenac	4'-Hydroxydiclofenac	Sulfaphenazole
CYP2C19	S-Mephenytoin	4'-Hydroxymephenytoin	Ticlopidine
CYP2D6	Dextromethorphan	Dextrorphan	Quinidine
CYP3A4	Midazolam	1'-Hydroxymidazolam	Ketoconazole

Positive controls are run as 7-point concentration responses on every run. Furafylline and ticlopidine (mechanism-based inhibitors) are preincubated with microsomes and NADPH for 15 minutes before probe addition. A run is accepted when each control IC₅₀ falls within 0.5-2 times the laboratory historical mean. Results are reported as mean percent inhibition relative to vehicle; values above 50% at 10 μ M are flagged, and small negative values indicate no inhibition.

5 Results

hERG results

All 38 compounds were tested. 1 of 38 compounds had a hERG IC₅₀ below 1 μ M. Most compounds (35 of 38) inhibited hERG with IC₅₀ below 10 μ M, so margins will depend on free therapeutic exposure. The median IC₅₀ was 3.8 μ M (1 below 1 μ M, 34 between 1 and 10 μ M, 3 at or above 10 μ M, of which 0 above the highest evaluable concentration). E-4031 gave an IC₅₀ of 11 nM in the accepted runs.

Compounds with QC notes: ST03-01224: First run rejected (E-4031 outside range); repeated; ST03-01366: First run rejected (E-4031 outside range); repeated. Values for these compounds should be read with the note.

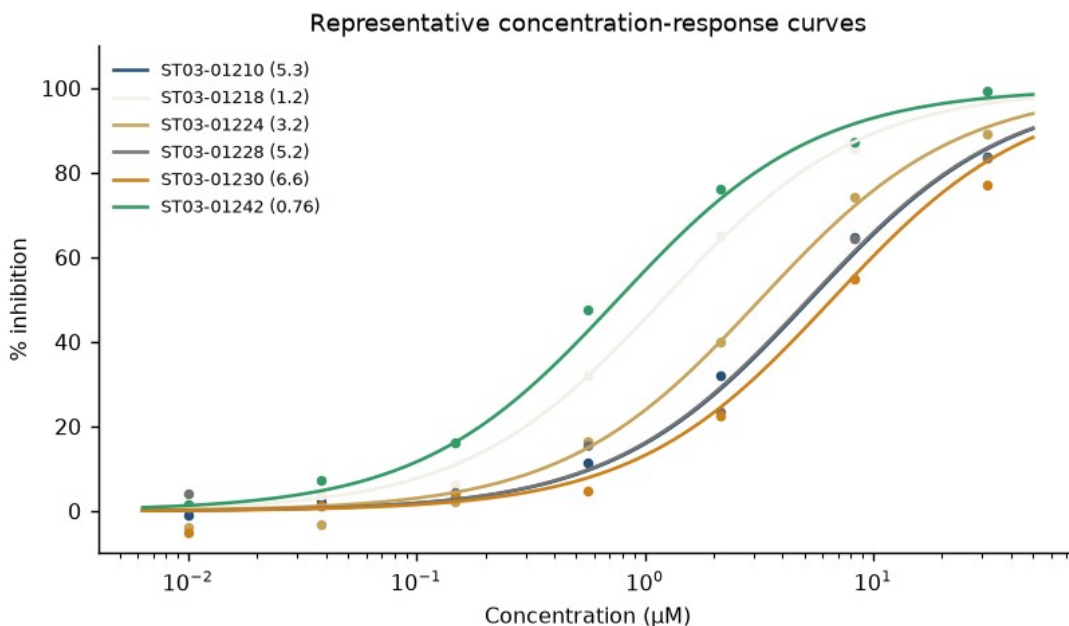
Table 2 hERG IC₅₀ by compound (μ M)

Compound	IC ₅₀ cell 1 (μ M)	IC ₅₀ cell 2 (μ M)	IC ₅₀ cell 3 (μ M)	IC ₅₀ geomean (μ M)	% inhibition @10 μ M	n cells (finite)	QC note
ST03-01210	5.3	5.2		5.2	59	2 (2)	
ST03-01218	1.2	1.6		1.4	88	2 (2)	
ST03-01224	3.2	3.0		3.1	73	2 (2)	First run rejected (E-4031 outside range); repeated
ST03-01228	5.2	2.7	3.0	3.5	75	3 (3)	
ST03-01230	6.6	16		10	52	2 (2)	
ST03-01242	0.76	1.2		0.97	97	2 (2)	
ST03-01246	26	21		24	26	2 (2)	
ST03-01252	2.0	1.5		1.7	88	2 (2)	
ST03-01254	3.9	6.1		4.9	69	2 (2)	

ST03-01264	1.8	1.5		1.7	89	2 (2)	
ST03-01266	2.7	3.1		2.9	76	2 (2)	
ST03-01270	4.5	8.1		6.0	62	2 (2)	
ST03-01276	5.7	5.1		5.4	66	2 (2)	
ST03-01278	0.72	1.4	1.1	1.1	93	3 (3)	
ST03-01282	9.6	3.1		5.4	58	2 (2)	
ST03-01288	3.8	4.0		3.9	71	2 (2)	
ST03-01290	4.0	3.1		3.5	73	2 (2)	
ST03-01294	5.5	5.0		5.3	60	2 (2)	
ST03-01296	8.5	11		9.8	47	2 (2)	
ST03-01300	4.8	3.8		4.2	75	2 (2)	
ST03-01306	5.1	5.0		5.1	62	2 (2)	
ST03-01308	3.9	3.8	2.7	3.5	75	3 (3)	
ST03-01312	2.9	3.3		3.1	80	2 (2)	
ST03-01314	1.4	1.2		1.3	92	2 (2)	
ST03-01320	3.7	4.1		3.9	74	2 (2)	
ST03-01324	4.8	5.3		5.0	64	2 (2)	
ST03-01326	1.3	1.3		1.3	98	2 (2)	
ST03-01330	3.1	3.0	2.7	2.9	77	3 (3)	
ST03-01338	4.1	4.5		4.3	71	2 (2)	
ST03-01354	3.6	3.4		3.5	70	2 (2)	
ST03-01356	1.4	1.8		1.6	80	2 (2)	
ST03-01360	3.5	4.0		3.7	72	2 (2)	
ST03-01362	1.9	2.1		2.0	80	2 (2)	
ST03-01366	7.2	4.7		5.8	56	2 (2)	First run rejected (E-4031 outside range); repeated
ST03-01372	16	9.2	13	12	49	3 (3)	
ST03-01380	3.8	5.1		4.4	69	2 (2)	
ST03-01384	2.2	2.7		2.4	85	2 (2)	
ST03-01386	4.8	5.4		5.1	70	2 (2)	

Top concentration 30 μ M; 6-point, 3-fold dilution; positive control E-4031 IC₅₀ 11 nM (acceptance 5-50 nM)

Compound IC₅₀ = geometric mean of finite cell IC₅₀s when they are the majority of accepted cells; otherwise reported as >30 μ M (or >10 μ M where solubility-limited)



CYP inhibition results

Inhibition above 50% at 10 μM was most frequent for CYP2C19. In total 17 compounds inhibited at least one isoform by more than 50% at 10 μM (CYP3A4: 10; CYP2D6: 4; CYP2C19: 13; CYP2C9: 8; CYP1A2: 10). All control IC₅₀s were within 0.5-2 times the historical mean (ketoconazole 0.027 μM against a historical mean of 0.030 μM).

Table 3 Percent inhibition at 10 μM

Compound	CYP1A2 (%)	CYP2C9 (%)	CYP2C19 (%)	CYP2D6 (%)	CYP3A4 (%)
ST03-01210	-1	9	8	3	-8
ST03-01218	37	1	8	21	31
ST03-01224	54	52	59	4	66
ST03-01228	-2	10	-0	-4	-1
ST03-01230	93	11	67	58	92
ST03-01242	100	38	63	34	91
ST03-01246	-4	5	-4	12	-1
ST03-01252	3	47	32	7	12
ST03-01254	7	2	5	-1	-2
ST03-01264	14	-1	-4	8	8
ST03-01266	36	-3	1	7	7
ST03-01270	7	0	2	-0	-2
ST03-01276	6	10	-3	-1	5
ST03-01278	55	-5	-1	7	5
ST03-01282	-9	27	51	1	6
ST03-01288	17	4	5	17	19
ST03-01290	62	86	63	5	38
ST03-01294	21	4	21	-9	30
ST03-01296	7	-7	-2	5	11
ST03-01300	3	6	1	-3	6
ST03-01306	40	40	58	7	14
ST03-01308	-3	8	29	0	19
ST03-01312	-6	-1	12	3	70
ST03-01314	91	56	73	54	90

ST03-01320	4	12	28	-2	5
ST03-01324	9	5	-3	-2	-3
ST03-01326	98	59	91	90	100
ST03-01330	37	62	60	-1	15
ST03-01338	71	62	89	0	78
ST03-01354	2	3	-1	12	42
ST03-01356	5	-0	1	8	15
ST03-01360	13	100	92	-4	43
ST03-01362	57	6	35	37	50
ST03-01366	-3	6	5	-7	-2
ST03-01372	12	8	-10	0	81
ST03-01380	12	70	66	0	61
ST03-01384	98	7	63	78	88
ST03-01386	4	6	3	-9	7

Single concentration 10 μ M, duplicate incubations, mean reported; negative values = no inhibition

Controls (IC₅₀, μ M): furafylline (CYP1A2) 1.9; sulfaphenazole (CYP2C9) 0.27; ticlopidine (CYP2C19) 1.5; quinidine (CYP2D6) 0.057; ketoconazole (CYP3A4) 0.027

Table 4 Positive control IC₅₀s

Control	Isoform	IC ₅₀ (μ M)	Historical mean (μ M)
Furafylline	CYP1A2	1.9	1.8
Sulfaphenazole	CYP2C9	0.27	0.35
Ticlopidine	CYP2C19	1.5	1.2
Quinidine	CYP2D6	0.057	0.060
Ketoconazole	CYP3A4	0.027	0.030

6 Deviations

The first hERG run for one compound was rejected because E-4031 fell outside its acceptance range; the compound was retested in a later run and only accepted data are reported.

7 Discussion

1 of 38 compounds had a hERG IC₅₀ below 1 μ M. Most compounds (35 of 38) inhibited hERG with IC₅₀ below 10 μ M, so margins will depend on free therapeutic exposure. Inhibition above 50% at 10 μ M was most frequent for CYP2C19. Values are in vitro and do not by themselves predict clinical risk; free exposure estimates are needed for that assessment.

8 Conclusion

Recommended next steps: prioritize compounds with hERG IC₅₀ at or above 10 μ M; for any compound below 10 μ M, compare the IC₅₀ with projected free plasma exposure before progression; repeat the flagged hERG measurements if a selection decision depends on them; run concentration responses for compounds flagged on CYP2C19 and any other isoform above 50%.

9 Records

Raw data, assay worksheets and instrument files are retained for 5 years under M15/BIO/06146.

Name	Role	Signature	Date
Matteo Rinalvesi	DMPK Lead	<u>M Rinalvesi</u>	December 28, 2015
Helen Marchwood	Bioanalytical Quality Manager	<u>H Marchwood</u>	December 27, 2015